

Introduction –Objective and FSI/Discretisation Strategy

Objective: We formulated, implemented & tested intriguing novel fluid-structure interaction (FSI) strategy for new wave-energy device with wave-buoy & buoy-generator interactions based on inequality constraints & Averaged Vector Field energy conservation. Used tedious bouncing ball model as pedagogically toy test. In five clear steps, our **novel discretisation strategy** is:

- **S1:** Write down the augmented Lagrangian with the inequality constraint embedded and derive the dynamics. **Why?** Includes all dynamics in one VP, directly with cunning coupling.
- **S2:** Approximate the Lagrange multiplier λ using new (implicit) function approximations to $\tilde{F}_+(\cdot)$, to solve for $\lambda(G) \leq 0$ in terms of constraint $G \geq 0$. **Why?** Dynamics is stiff & geometric integrators perform poorly with (well-)known explicit functions $\lambda(G)$.
- **S3:** Write down new Lagrangian/Hamiltonian dynamics with λ eliminated. **Why?** Simpler & smoothed.
- **S4:** Use Average Vector Field energy (AVF) conservation as time integrator. **Why?** Working best to date. Symplectic integration failed.
- **S5:** Test! Derive/Obtain nonlinear solver stability criterion for Δt . **Why?** Get a handle on time-step size & convergence.

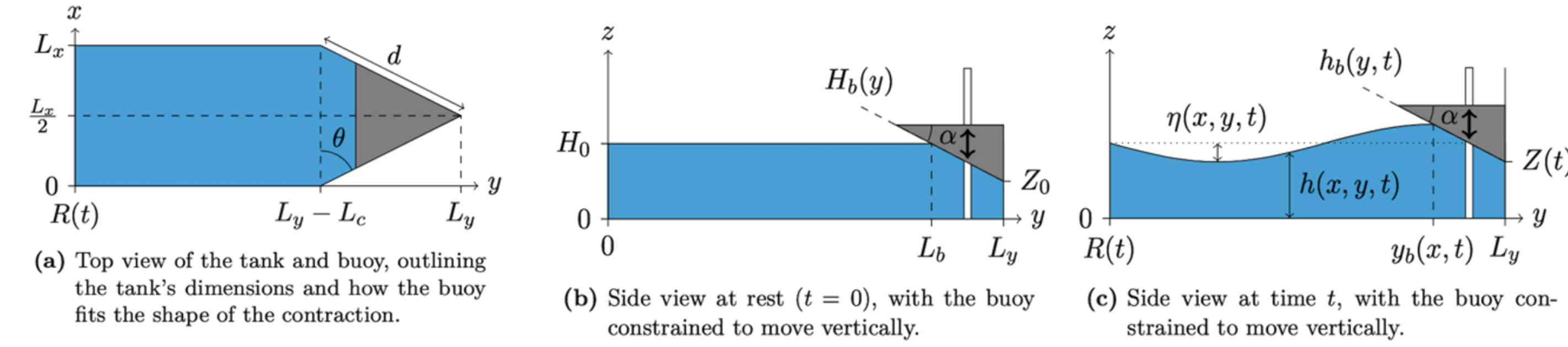


Figure 1. Sketches wave-energy device, buoy $h_B(Z(t), y)$. Courtesy Jonathon Bolton (dissertation in preparation).

S1,2,3: Models with inequality constraints

S1: Consider **bouncing ball** moving in vertical under constant gravity above $Z(t) \geq 0$, time t . It **bounces** with instant momentum reversal off the table. Augmented variational principle (VP) [2]:

$$0 = \delta \mathcal{L}_B[Z, W, \lambda] \equiv \delta \int_0^T W \dot{Z} - \frac{1}{2} W^2 - Z - \frac{1}{2\gamma} \left(F_+(-\gamma Z - \lambda)^2 - \lambda^2 \right) dt \quad (1)$$

with $\dot{Z} \equiv dZ/dt$, particle velocity $W(t)$, Lagrange multiplier $\lambda(t)$, constant $\gamma \gg 0$, and (maximum) function $F_+(Q) \equiv \max(0, Q)$ with $-Z \leq 0$. Karush-Kuhn-Tucker (KKT) inequalities result from (2b): $-Z \leq 0$, $\lambda \leq 0$, $\lambda G(Z) = \lambda Z = 0$. Variations of (1) subject to $\delta Z(0) = \delta Z(T) = 0$ yield:

$$\delta W : \dot{Z} = W = \frac{\partial H}{\partial W}, \quad \delta Z : \dot{W} = -\frac{\partial H}{\partial Z} = -1 + \tilde{F}_+(-\gamma Z - \lambda) = -1 - \lambda \quad (2a)$$

$$\delta \lambda : \lambda = -\tilde{F}_+(-\gamma Z - \lambda) \quad \text{with} \quad \tilde{F}_+(Q) = F_+(Q)F'_+(Q) = \max(0, Q). \quad (2b)$$

S2: Seek implicit and explicit approximations of $\lambda, \tilde{F}(Q), \tilde{F}'(Q)$ to enforce stability ($0 < b, d \ll 1$):

$$\text{Smooth} : \lambda_1(Z) \approx -ae^{-\gamma Z/b} = -\tilde{F}_+(Q) = -\gamma Z - \lambda, Q = b \ln(\tilde{F}_+(Q)/a) + \tilde{F}_+(Q) \quad (3a)$$

$$\text{Cut-off} : \lambda_2(Z) \approx \min(0, \frac{Z}{b}), \tilde{F}_+(Q) = \frac{\max(0, Q)}{1 + b\gamma}, \quad F_+(Q) = \frac{1}{\sqrt{1 + b\gamma}} \max(0, Q) \quad (3b)$$

$$\text{Asymp.} : \lambda_3(Z) \approx a \min\left(0, \frac{Z}{b} + \frac{|Z|}{d}\right), \tilde{F}_+(Q) = \max\left(0, Q + \frac{(d\Gamma - d\sqrt{\Gamma^2 + 4aQ/d})}{2a/\gamma}\right) \quad (3c)$$

$$C_2 : \lambda_4(Z) \approx a \left\{ \frac{Z}{b}, P(Z; \delta), 0 \right\} \quad \text{with} \quad P(Z) \in C_2\text{-spline}, \tilde{F}(Q), F(Q) \text{ implicit}, \quad (3d)$$

with regions $\{Z \leq -\delta, |Z| < \delta, Z > \delta\}$. **S3:** The integral of these forces $\lambda_i(Z)$ ($i = 1, 2, 3, 4$) with respect to constraint variable $G = Z$ yields a potential $V_\lambda(G)$ in the augmented Lagrangian (1).

Wave-energy model

S1,S2,S3: Variational Boussinesq Model (VBM) water waves with surface velocity potential $\phi(x, y, t) = \phi_s(x, y, t)$ & depth $h(x, y, t)$ **coupled** to heaving buoy and power generator:

$$0 = \rho_0 \delta \int_0^T \int_0^{L_x} \int_{R(t)}^{l_y(x)} \phi \partial_t h \, dy \, dx + MW \dot{Z} + (L_i I - K(Z)) \dot{Q} - \mathcal{H} \, dt \quad (4)$$

$$\begin{aligned} &\equiv \rho_0 \delta \int_0^T \int_0^{L_x} \int_{R(t)}^{l_y(x)} \phi \partial_t h - \left(\frac{1}{2} h |\nabla \phi + h \psi \nabla h + \frac{1}{3} h^2 \nabla \psi|^2 + \frac{1}{6} h^3 \psi^2 + \frac{\beta}{90} h^5 |\nabla \psi|^2 \right) \\ &\quad - gh \left(\frac{1}{2} h - H_0 \right) - \frac{1}{2\gamma} \left(F_+(\gamma(h - h_b) - \lambda)^2 - \lambda^2 \right) dy \, dx - \int_0^{L_x} \dot{R} \left(h\phi + \frac{1}{3} h^3 \psi \right) |_{y=R} dx \\ &\quad + MW \dot{Z} - \frac{1}{2} MW^2 - MgZ + (L_i I - K(Z)) \dot{Q} - \frac{1}{2} L_i I^2 dt, \end{aligned} \quad (5)$$

via constraint $G = (h_b - h) \geq 0$, with $\mathcal{H}$ total energy, $\beta = 1$ for VBM and $\beta = 0$ for Green-Naghdi model [4]. Variation of this VBM-VP with respect to variables $\mathbf{q} \equiv \{h, \phi, \psi, Z, W, Q, I\}$ yields the system of equations for (conservative part of) the entire wave-energy device ([1]):

$$\delta \phi : \quad \partial_t h = \frac{1}{\rho_0} \frac{\delta \mathcal{H}}{\delta \phi}, \quad \delta \psi : \quad 0 = -\frac{1}{\rho_0} \frac{\delta \mathcal{H}}{\delta \psi}, \quad \delta \lambda : \quad 0 = \frac{1}{\rho_0} \frac{\delta \mathcal{H}}{\delta \lambda}, \quad \delta h : \quad \partial_t \phi = -\frac{1}{\rho_0} \frac{\delta \mathcal{H}}{\delta h} = -B + \lambda \approx -B + \underline{\lambda}_i \quad (6a)$$

$$\delta W : \quad \dot{Z} = \frac{1}{M} \frac{\partial \mathcal{H}}{\partial W} = W \quad (6b)$$

$$\begin{aligned} \delta Z : \quad \dot{W} + \frac{\gamma_m}{M} G_I(Z) \dot{Q} &= -\frac{1}{M} \frac{\partial \mathcal{H}}{\partial Z} = -g + \frac{\rho_0}{M} \int_0^{L_x} \int_{R(t)}^{l_y(x)} F_+(\gamma(h - h_b)) F'_+(\gamma(h - h_b)) \, dy \, dx \\ &= -g - \frac{\rho_0}{M} \int_0^{L_x} \int_{R(t)}^{l_y(x)} \lambda \, dy \, dx \approx -g - \frac{\rho_0}{M} \int_0^{L_x} \int_{R(t)}^{l_y(x)} \underline{\lambda}_i \, dy \, dx \end{aligned} \quad (6c)$$

$$\delta I : \quad \dot{Q} = \frac{1}{L_i} \frac{\partial \mathcal{H}}{\partial I} = I, \quad \delta Q : \quad \dot{I} - \frac{\gamma_m}{L_i} G_I(Z) \dot{Z} = -\frac{1}{L_i} \frac{\partial \mathcal{H}}{\partial Q} - \text{dissipation} \equiv -\frac{(R_i + R_c)}{L_i} I - \frac{I}{L_i |I|} V_s(|I|), \quad (6d)$$

with Bernoulli term B , added **electrical resistance/harvesting load** & **λ_i -approximation** (function of $h_B(Z(t), y) - h(x, y, t)$, cf. (3)). Nonlinear **$K'(Z) \equiv G_I(Z)$** solves axi-symmetric 3D Maxwell's equations [1].

S5 testing & results: bouncing ball

Using $\lambda_3(Z)$ (3c) gives better convergence, since $F_+(Q) \rightarrow Q$ as $Q \rightarrow \infty$. Error is $O(\Delta t)^p$ with $p \approx 1$ compared with exact solution at $t \approx 3.5T$ for $a = 10, b = d = 0.1\Delta t^2, 10\Delta t^3$ (set-1, set-2).

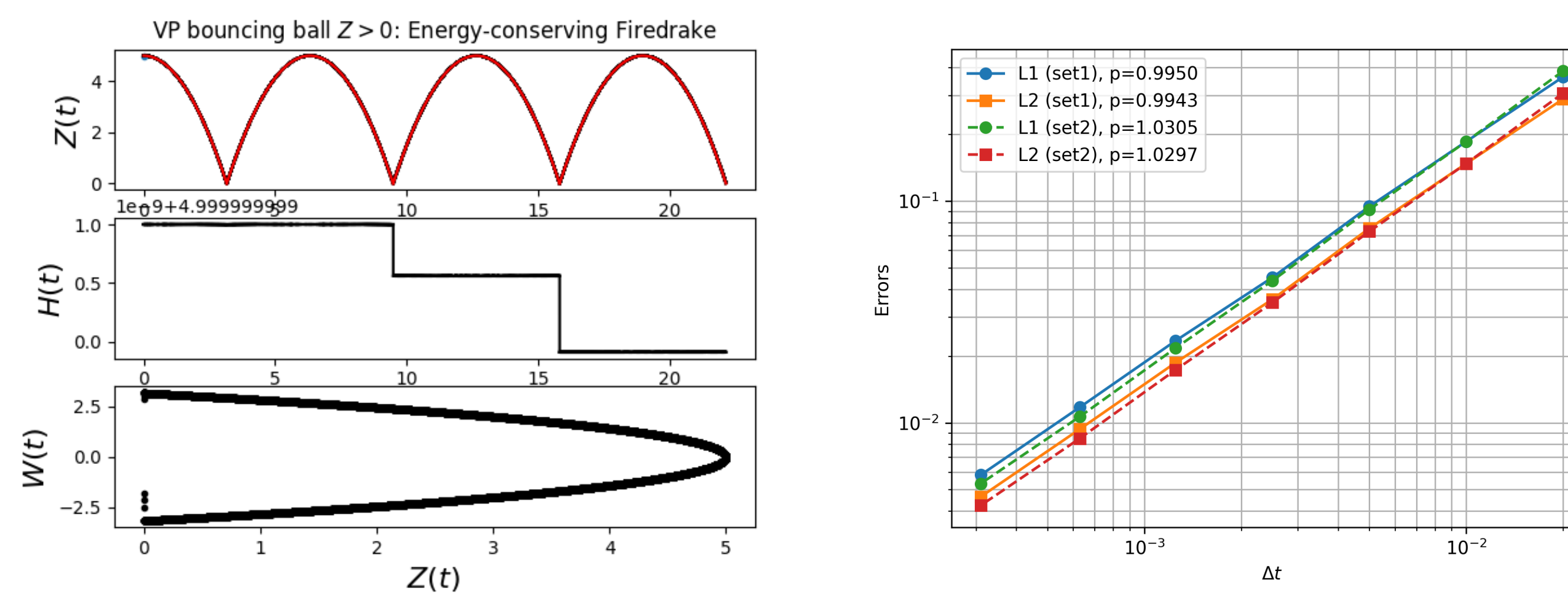


Figure 2. Left: Shown is ball under gravity bouncing off floor at $Z = 0$. Δt is not tuned. Position vs. time (exact and numerical solutions shown), energy $H(t)$ fluctuation $\sim 10^{-9}$ around $H_0 = 5$ and phase diagram $\{Z^n, W^n\}$ of numerics. Right: errors; finest errors are L_1 -norm: (0.00582, 0.00532) & L_2 -norm: (0.00464, 0.00424) at $\Delta t = 0.00031$.

Singularities in AVF have to be dealt with exactly for stability & accuracy, i.e. in Newton iterations:

$$\frac{(W^{n+1} - W^n)}{\Delta t} = -1 - a \left(\frac{(Z_-^{n+1} + Z_-^n)}{2b} - \frac{(Z_-^{n+1} + Z_-^n + Z_-^{n+2})}{3d} \right) \begin{cases} 1 & Z_-^{n+1} = Z_-^n \\ \frac{Z_-^{n+1} - Z_-^n}{Z_-^{n+1} - Z_-^n} & \text{else} \end{cases}$$

S4 AVF energy-conservation strategy

Spectrally accurate Continuous Galerkin (CG) finite element methods were used in space (e.g., 2nd order CG1, CG2, CG3) with time-efficient & error-reducing automation via & implemented in *Firedrake* environment [5]. **Key take-away** is that energy conserving 2nd-order AVF discretisation had to be used for stability, over time $t \in [t^n, t^{n+1} = t^n + \Delta t]$. Hence, we use linear-in- s variables $\mathbf{q}(s) \equiv (Z(s) = (Z^n + s(Z^{n+1} - Z^n), W(s)))$ & likewise for $\mathbf{q}(s) \equiv \{h(x, y, s), \phi(x, y, s), \psi(x, y, s), Z(s), W(s), I(s)\}$ (e.g., [3]):

$$\frac{\partial \mathbf{q}}{\partial t} = \frac{1}{\Delta t} \frac{d\mathbf{q}}{ds} = J(\mathbf{q}^{n+1/2}) \int_0^1 \frac{\partial H[\mathbf{q}(s)]}{\partial \mathbf{q}(s)} ds, \quad (7)$$

with **crucial exact integration**. For polynomials this is automated in Firedrake and for the λ_i 's this is derived/programmed due to difficult switches & limit $\mathbf{q}^{n+1} = \mathbf{q}^n$ in Newton iterations. J is skew-symmetric tensor for (2) (J constant) or (6) (J depends on $G_I(Z)$ in W, I -equations).

S5 Testing & results: wave-energy device

For our device, $\lambda_1(G)$ works **poorly** since exponential explodes and, hence, $\lambda_4(G)$ is **best** (using $a = g, b = 10\Delta t^{3/2}$). Crucial C_2 smoothness in $\lambda_{1,4}$ is needed in CG-space. λ_4 is still too noisy.

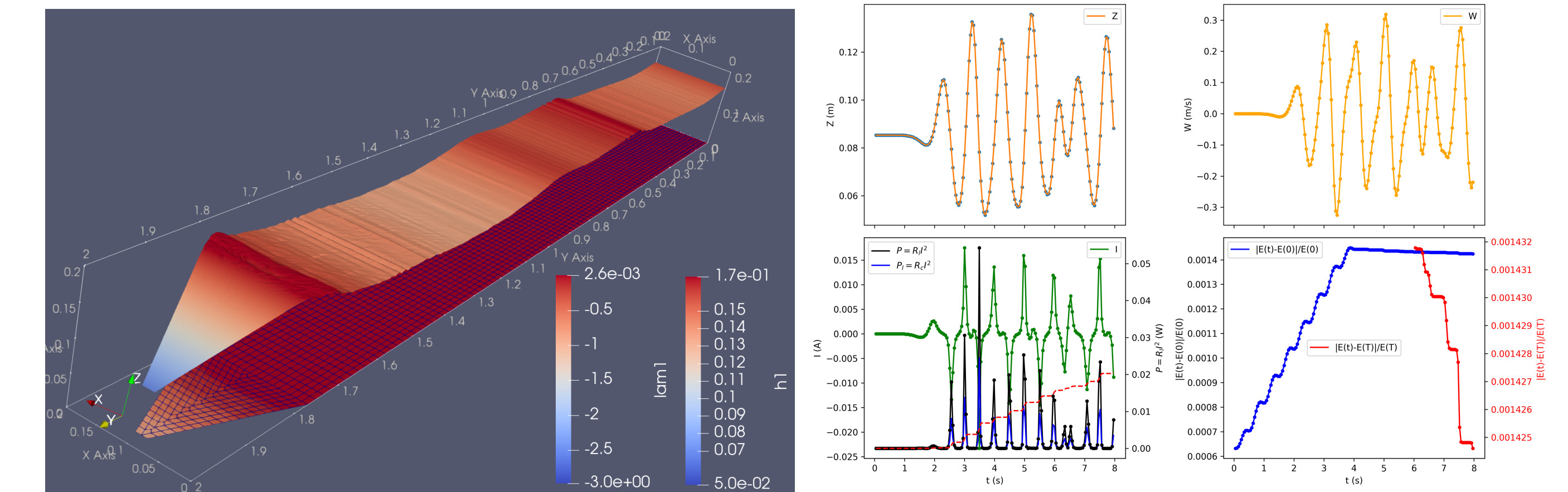


Figure 3. Left: snapshot free surface $h(x, y, t)$ & $\lambda_4(x, y, t)$ at $t = 4.08s$ on mesh. Right: $Z(t), W(t), I(t)$ & harvest/circuit dissipation & scaled $H(t)$ vs t . CG2 run. Post-processed λ_i still too noisy, but buoy motion & generator dynamics stable & smooth. Stable CG1–CG3 simulations completed. NB: $\delta \approx \Delta x$; buoy not shown, only its imprint.

Conclusions

The **key take-aways** of our new FSI-AVF framework for our new wave-energy device are:

- that singularities arising in AVF-integration, especially around the water- or contact line, need to be dealt with exactly; instant or continuous contact (C_2 needed) cases differ; and,
- that the free surface region needs to be solved exactly, i.e. have $\lambda = 0$; both work in progress.

References

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